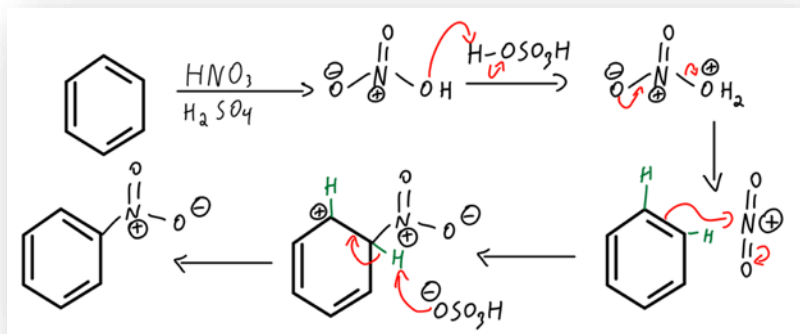
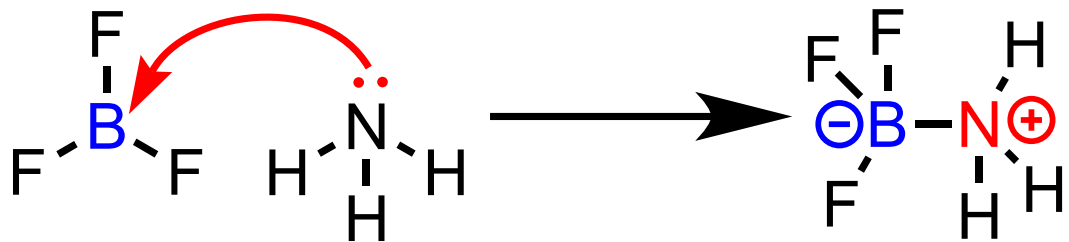
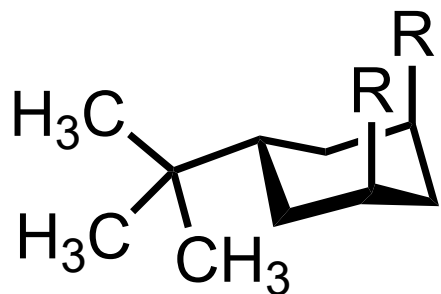
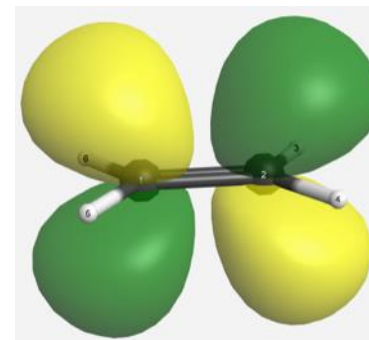




Fundamentals of Organic Reaction Mechanisms



Organic Chemistry II
CHEM 2312



Learning Objectives

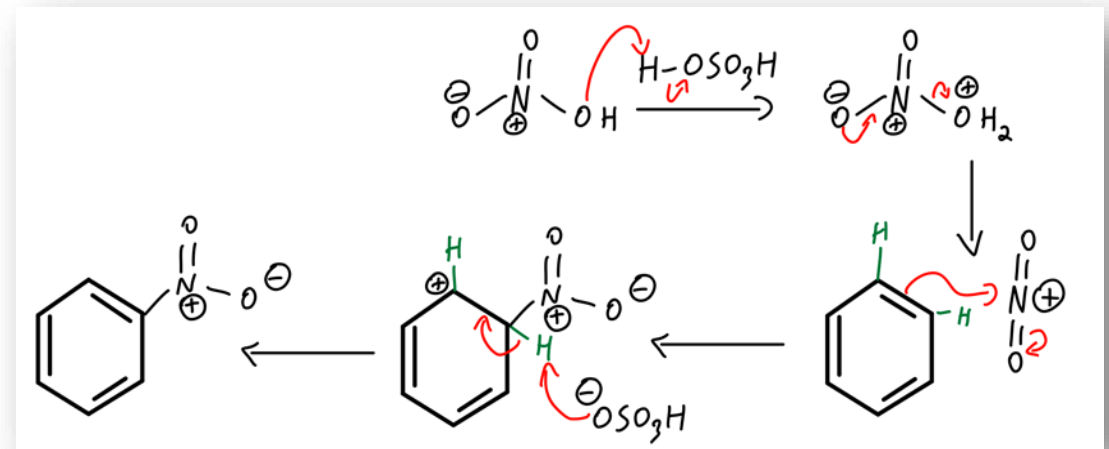
- By the end of this module, you should be able to:
 - Deduce kinetic and thermodynamic information from reaction coordinate diagrams
 - Draw curved arrows given the reactants and products of an elementary step; deduce products from reactants and curved arrows
 - Connect curved arrows to overlap between localized molecular orbitals
 - Recognize and draw the elementary steps of organic reaction mechanisms
 - Predict the favored side of an acid-base equilibrium using pK_a values
 - Judge the relative stabilities of two or more molecules using the five structural factors that affect stability

From the outset you will need intimate familiarity with the principles of kinetics, thermodynamics, and equilibrium from CHEM 1211K and 1212K.

Reaction Mechanisms | 1 of 3

A **reaction mechanism** is a series of elementary steps showing how a set of reactants transforms into a set of products (usually through one or more **intermediates**).

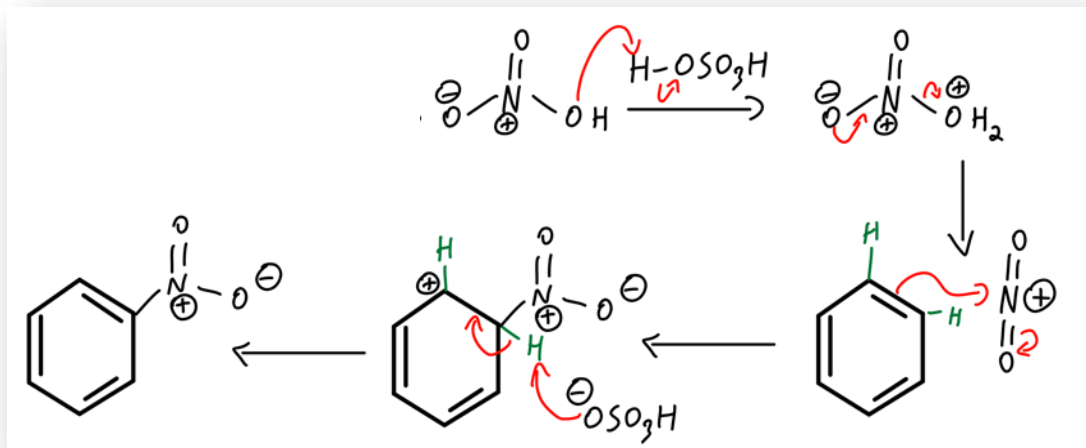
- Explains the dependence of the reaction on various experimental variables
- Suggests strategies for accelerating the reaction
- Suggests analogous reactants for related reactions
- Provides evidence for the generality of elementary steps



Reaction Mechanisms | 2 of 3

A **reaction mechanism** is a series of elementary steps showing how a set of reactants transforms into a set of products (usually through one or more **intermediates**).

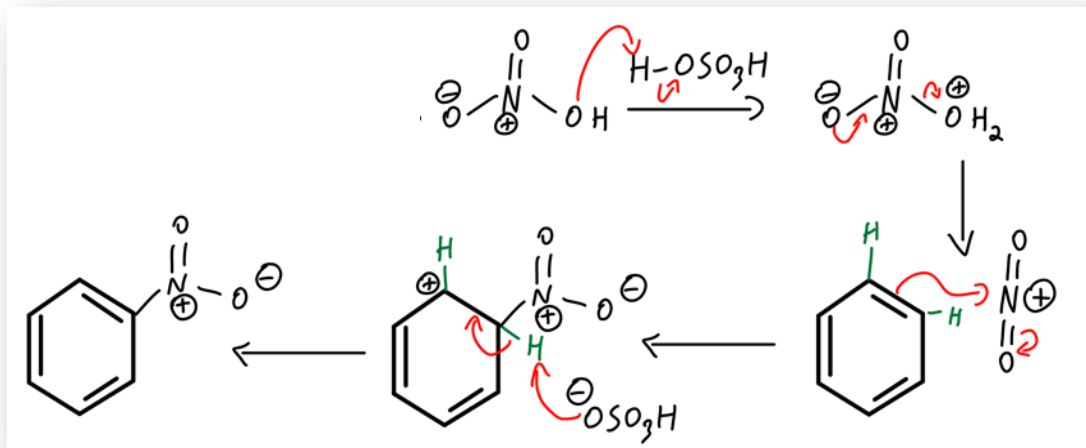
- **Elementary step:** a single molecular collision involving conversion of reactants to products through one transition-state structure
 - For our purposes, involves one orbital interaction or a pair of complementary interactions
 - Limited to about ten fundamental types
 - Kinetics depends on molecularity of the step



Reaction Mechanisms | 3 of 3

A **reaction mechanism** is a series of elementary steps showing how a set of reactants transforms into a set of products (usually through one or more **intermediates**).

- **Intermediate:** a stable species (energy minimum) between the reactants and products in a multi-step mechanism
- **Transition state:** an energy maximum that occurs between the initial and final structures of an elementary step.



Equations and Concepts of Thermodynamics

$$\Delta G^{\circ} = \Delta H^{\circ} - T\Delta S^{\circ}$$

The free energy change of a process is a temperature-dependent balance between enthalpy and entropy changes.

Enthalpy reflects the strengths of bonds and intermolecular forces.

Entropy reflects the extent of disorganization and energy dispersal.

$$\Delta G^{\circ} = -RT \ln K$$

The concentrations of reactants and products at equilibrium, as captured in K , are exponentially related to ΔG° .

You should be able to predict the signs of ΔH° and ΔS° for a reaction or step given the reactants and products.

Equations and Concepts of Kinetics

- Dependence of reaction rate on temperature and concentration is built into the Arrhenius equation and rate law

$$k = Ae^{-\Delta G^\ddagger / RT} \quad \text{rate} = k[A]^m[B]^n$$

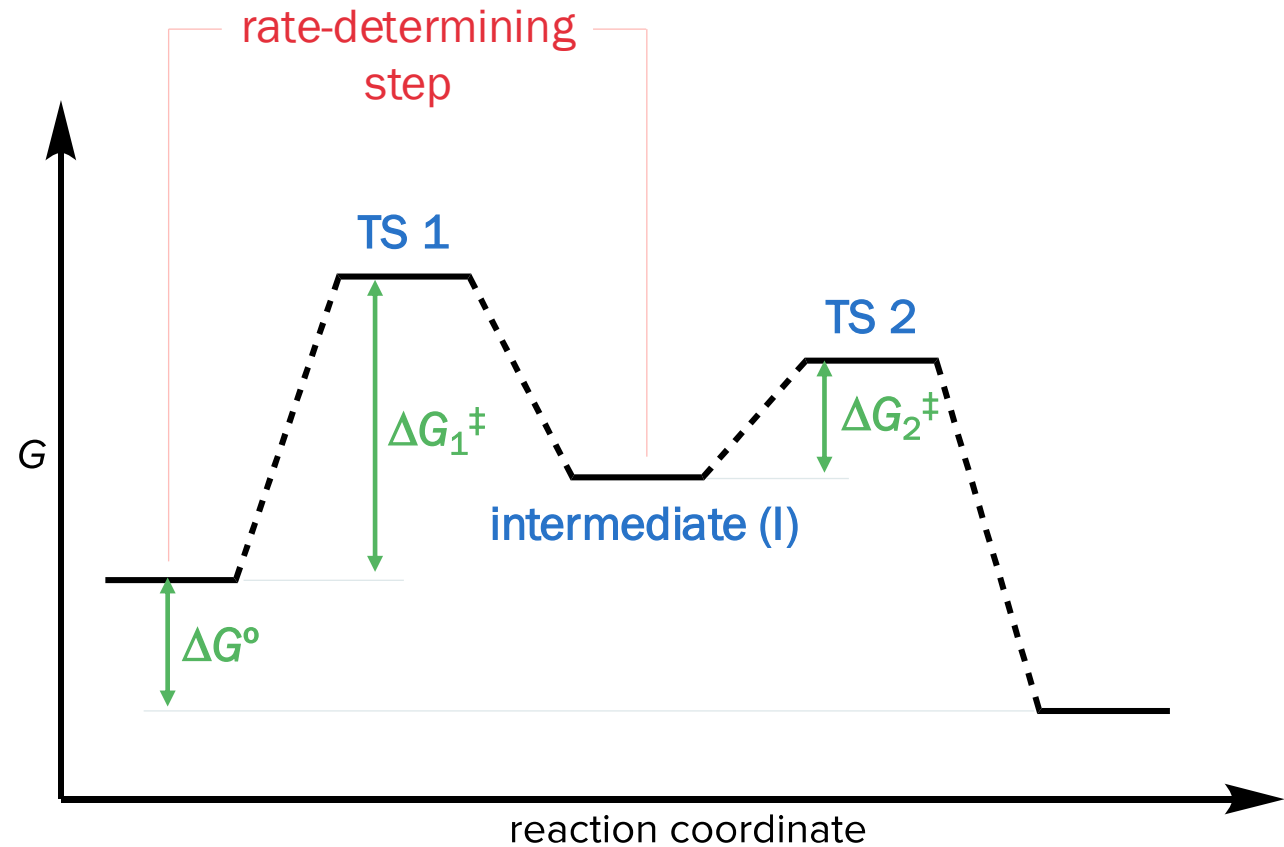
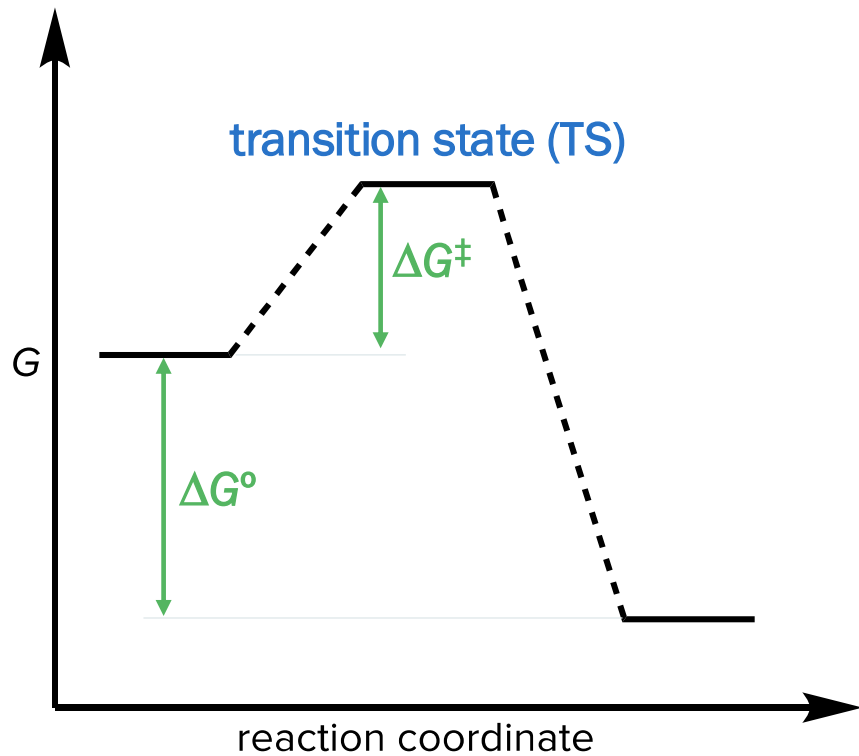
↑
Activation energy

- Most mechanisms we will study have a step that is significantly slower than the rest: the rate-determining step (RDS)
- Observed rate law reflects the molecularity (number of molecules reacting) of the RDS

The form of the rate law depends on the reaction mechanism and thus, it can be used to rule out physically impossible mechanisms.

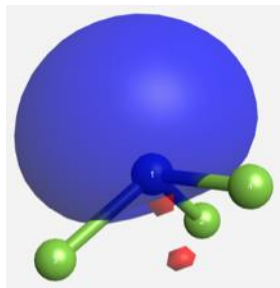
Reaction Coordinate Diagrams

A *reaction coordinate diagram* charts the energetic course of a reaction mechanism.

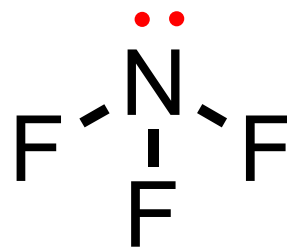


Localized Molecular Orbital Theory

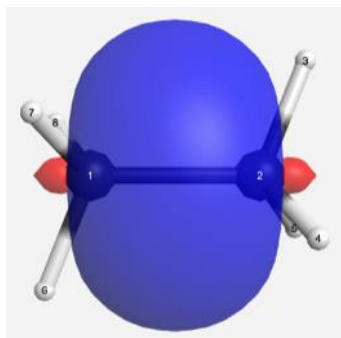
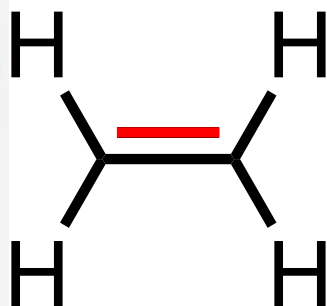
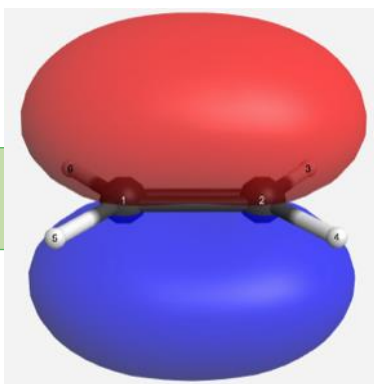
Electron Sources



nonbonding lone pair (n)



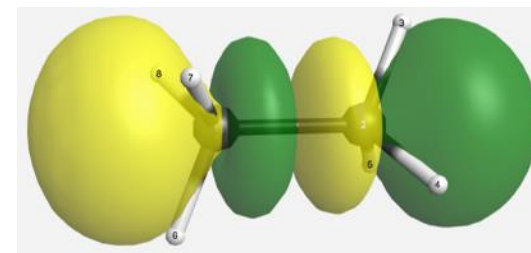
pi bond (π)



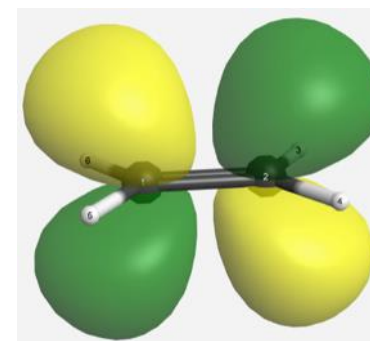
sigma bond (σ)



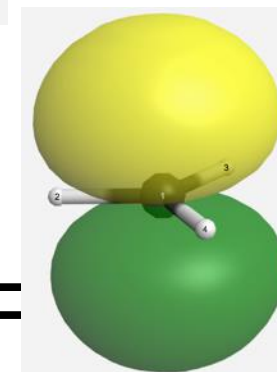
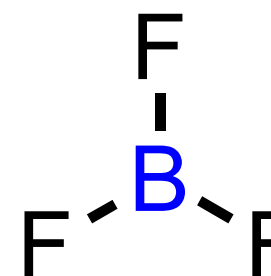
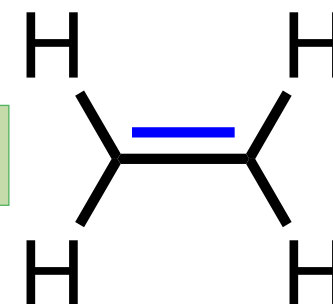
Electron Sinks



$\text{H}_3\text{C}-\text{CH}_3$
sigma antibond (σ^*)



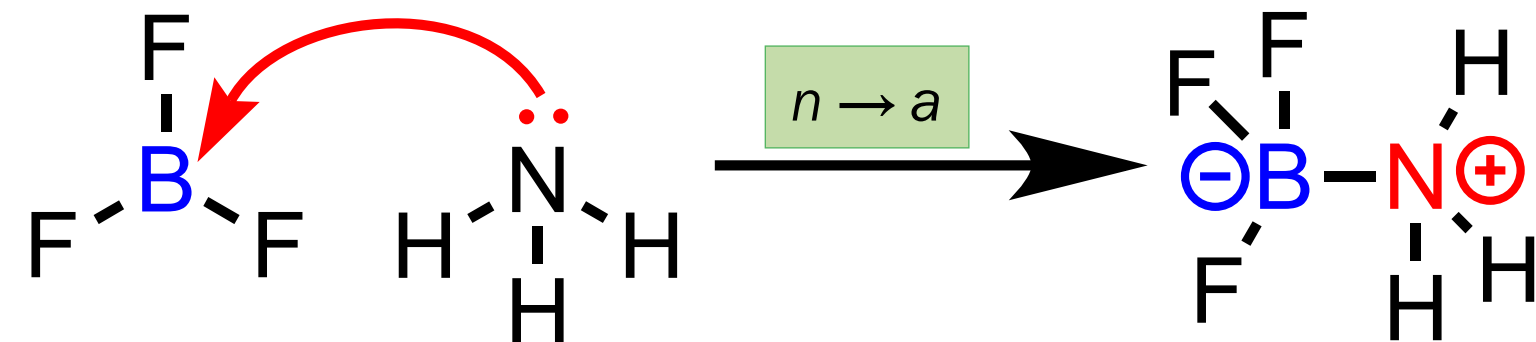
pi antibond (π^*)



empty $2p$ orbital (a)

Reaction Mechanisms and Curved Arrows

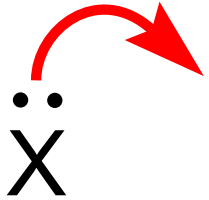
- An **elementary step** involves a single interaction between two localized molecular orbitals, one filled and one empty
- **Curved arrows** show how electrons move during elementary steps of organic reaction mechanisms
 - A curved arrow starts at an **electron source** and ends at an **electron sink**
 - Each arrow represents the movement of *two electrons*
 - Each arrow generates either a bond or a lone pair



Note the changes in formal charge at the source and sink atoms. N has formally lost one electron; B has formally gained one.

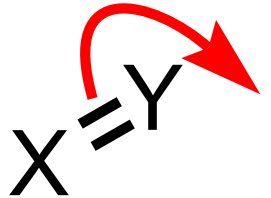
Electron Sources and Sinks | 1 of 2

There are three and only three classes of electron sources.
All electron flow must start at one of the following sources.



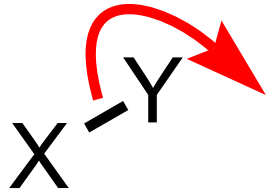
nonbonding lone pair (n)

A lone pair is donated, generating a new bond at X.



pi bond (π)

A pair of π electrons is donated, generating a new bond at Y and increasing the charge by 1 at X. An X-Y single bond remains.

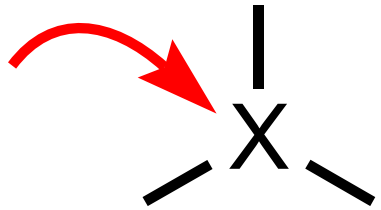


sigma bond (σ)

A pair of σ electrons is donated, generating a new bond at Y and increasing the charge by 1 at X. The X-Y bond is cleaved.

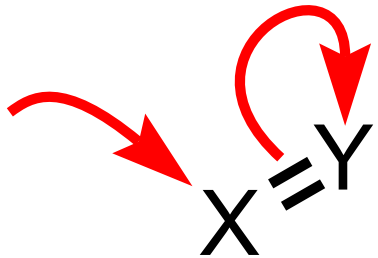
Electron Sources and Sinks | 2 of 2

There are three and only three classes of electron sinks.
All electron flow must end at one of the following sinks.



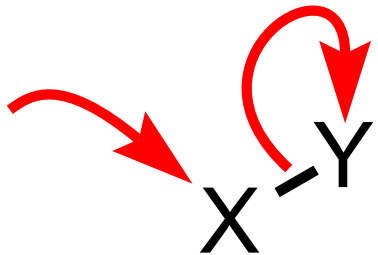
empty $2p$ atomic orbital (a)

A pair lands at X, forming a new bond to X and decreasing its charge by 1.



pi antibond (π^*)

A pair lands at X, pushing the π electrons to Y (generating a lone pair) and decreasing the charge of Y by 1. An X-Y single bond remains.

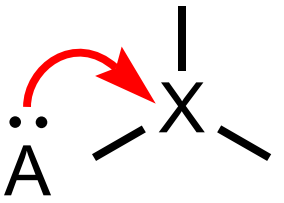
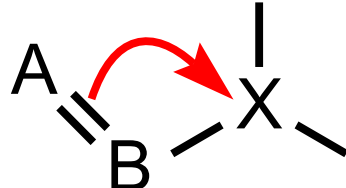
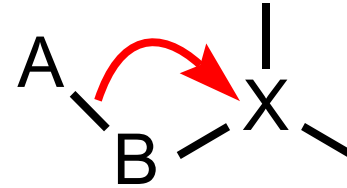
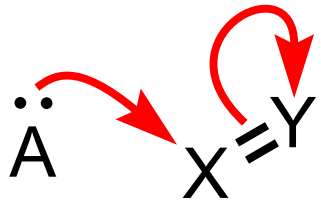
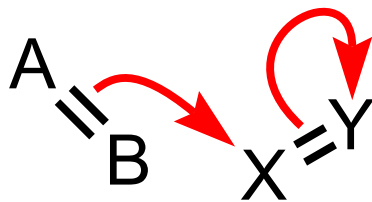
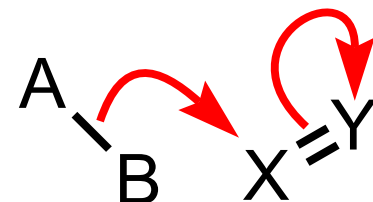
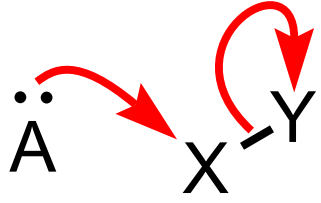
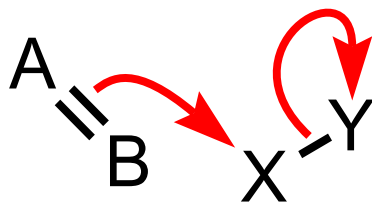
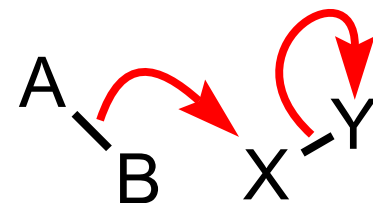


sigma antibond (σ^*)

A pair lands at X, pushing the s electrons to the Y (generating a lone pair) and decreasing the charge of Y by 1. The X-Y bond is cleaved.

Anatomy of an Elementary Step | 1 of 2

There are ten elementary steps of polar organic reaction mechanisms. Each can be described by curved arrows and a unique localized MO interaction. *Curved arrows imply the localized MOs involved.*

	Nonbonding pair (n)	Pi bond (π)	Sigma bond (σ)
Empty atomic orbital (a)			
Pi antibond (π^*)			
Sigma antibond (σ^*)			

Anatomy of an Elementary Step | 2 of 2

There are ten elementary steps of polar organic reaction mechanisms. Each can be described by curved arrows and a unique localized MO interaction. *Curved arrows imply the localized MOs involved.*

	Nonbonding pair (n)	Pi bond (π)	Sigma bond (σ)
Empty atomic orbital (a)	$n \rightarrow a$	$\pi \rightarrow a$	$\sigma \rightarrow a$
Pi antibond (π^*)	$n \rightarrow \pi^*$	$\pi \rightarrow \pi^*$	$\sigma \rightarrow \pi^*$
Sigma antibond (σ^*)	$n \rightarrow \sigma^*$	$\pi \rightarrow \sigma^*$	$\sigma \rightarrow \sigma^*$

Overview of Common Elementary Steps

1. Proton transfer (pt) $n \rightarrow \sigma^*$
2. Dissociation of a nucleofuge (D_N) $\sigma \rightarrow a$
3. Association of a nucleophile (A_N) $n \rightarrow a$
4. Dissociation of an electrofuge (D_E) $\sigma \rightarrow a$
5. Association of an electrophile (A_E) $\square \rightarrow a$
6. 1,2-Rearrangement (1,2R) $\sigma \rightarrow a$
7. Nucleophilic addition to a π bond (Ad_N) $n \rightarrow \pi^*$
8. Beta-elimination (E_β) $n \rightarrow \sigma^*$
9. Bimolecular elimination (E2) $\sigma \rightarrow \sigma^*$
10. Bimolecular nucleophilic substitution (S_N2) $n \rightarrow \sigma^*$

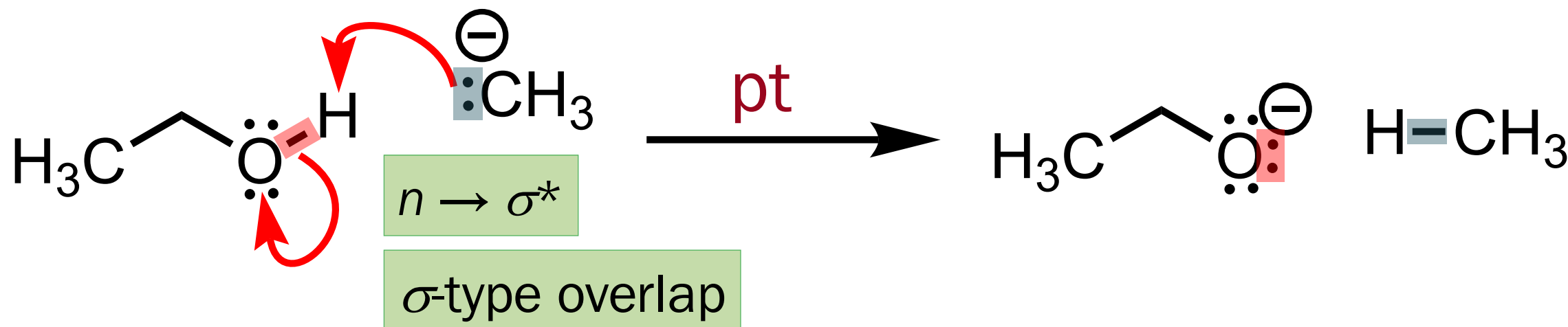
Nucleofuge: the group that departs with a pair of electrons when a bond breaks heterolytically.

Electrofuge: the group that ends up with no electrons when a bond breaks heterolytically.

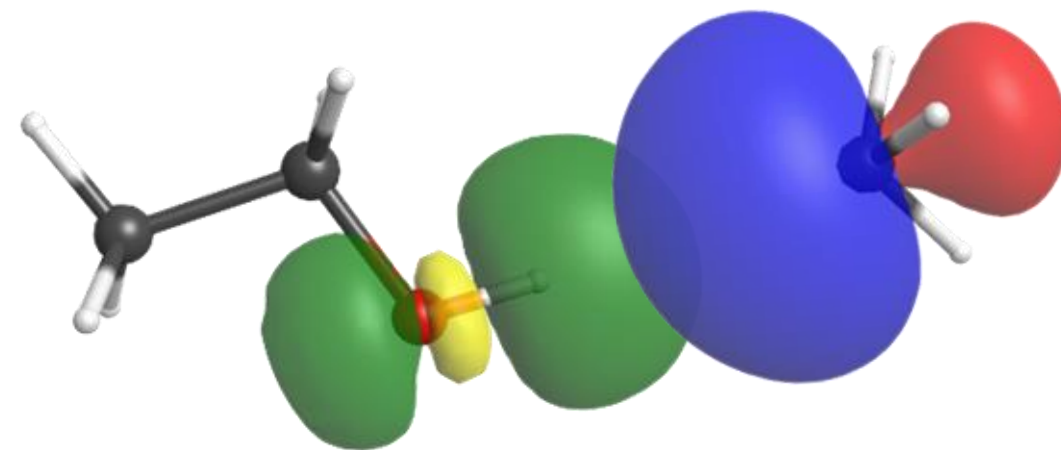
Proton Transfer (pt)



Acid Base Reactions 11



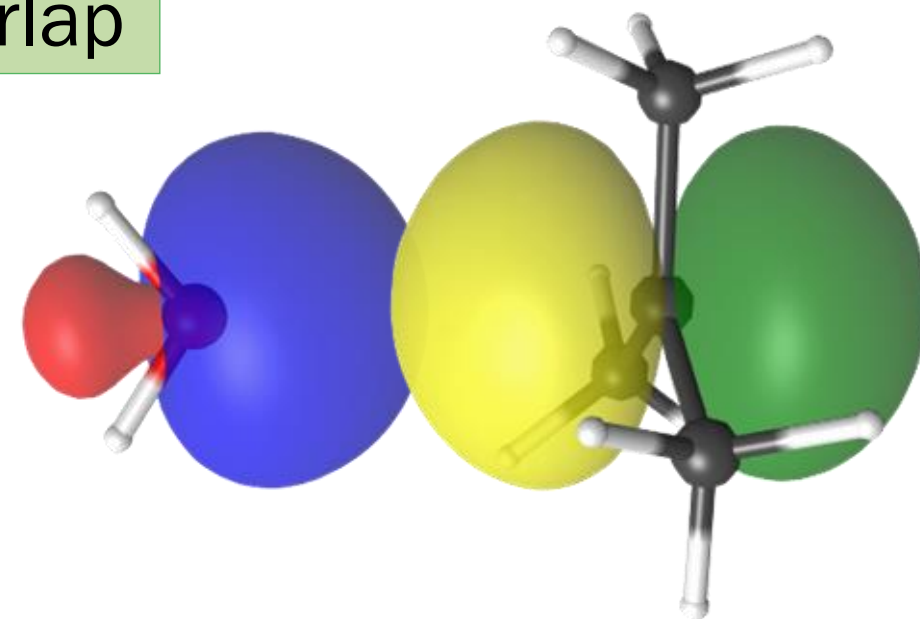
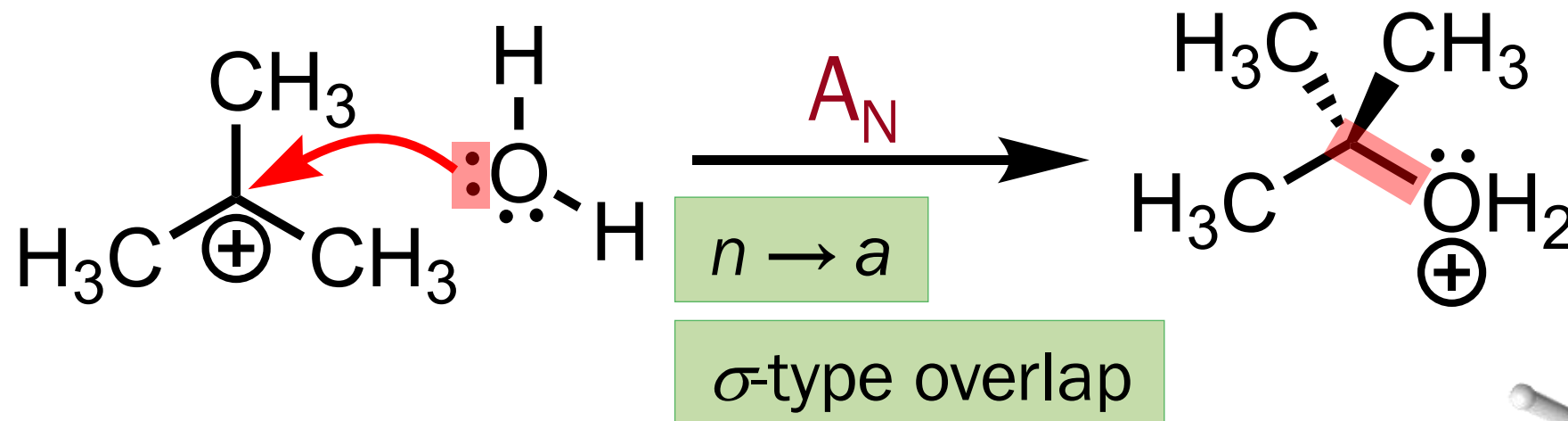
Curved arrows imply the localized molecular orbitals involved in an elementary step. In proton transfer, a nonbonding lone pair on the base B overlaps with the σ^* orbital of the H-A bond. The products are the conjugate acid of the base (HB^+) and the conjugate base of the acid (A^-). Proton transfers are *always* thermodynamically controlled (driven by $\text{p}K_a$ differences).



Association of a Nucleophile (A_N)



Substitution Reactions 3

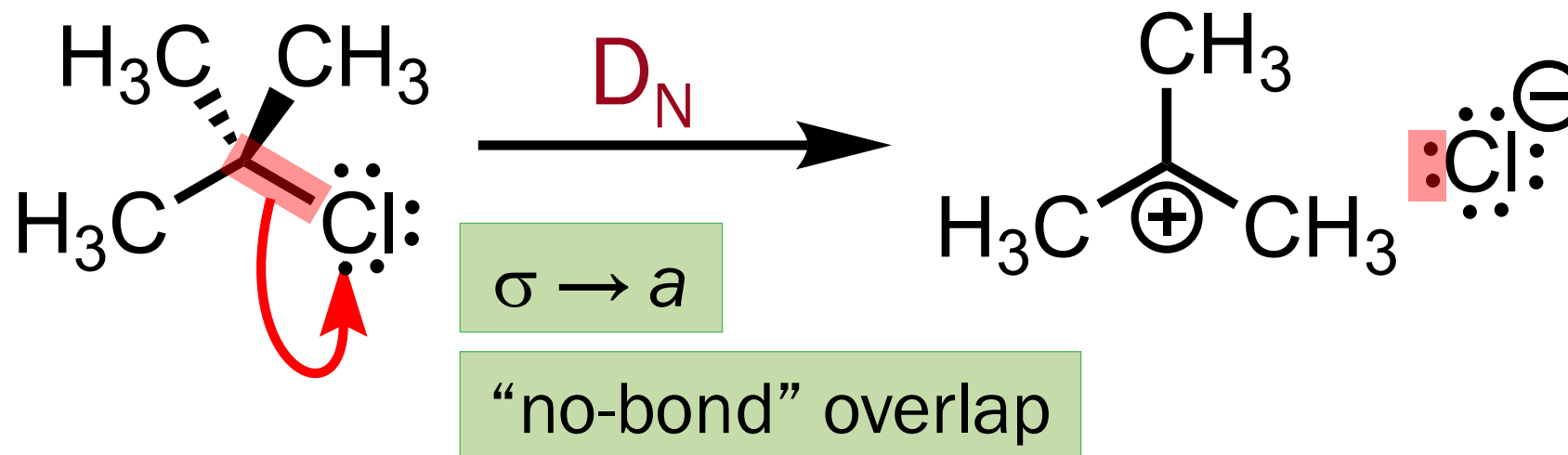


Curved arrows imply the localized molecular orbitals involved in an elementary step. In A_N , a nonbonding lone pair on the nucleophile overlaps with an empty atomic orbital on the electrophile. The product contains a new sigma bond between the nucleophilic and electrophilic atoms. The charge on the nucleophile increases by 1 and the charge on the electrophile decreases by 1.

Dissociation of a Nucleofuge ($\underline{D_N}$)



Substitution Reactions 3

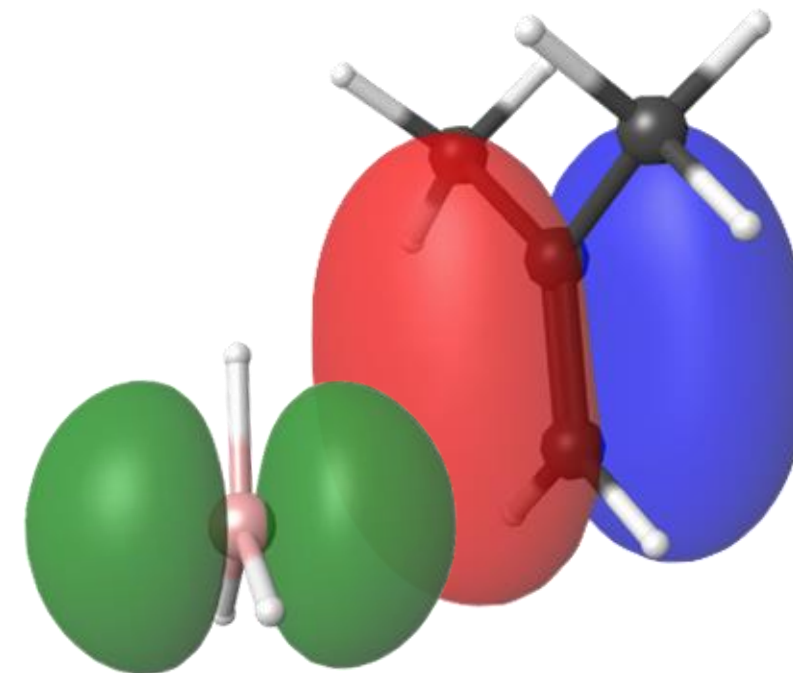
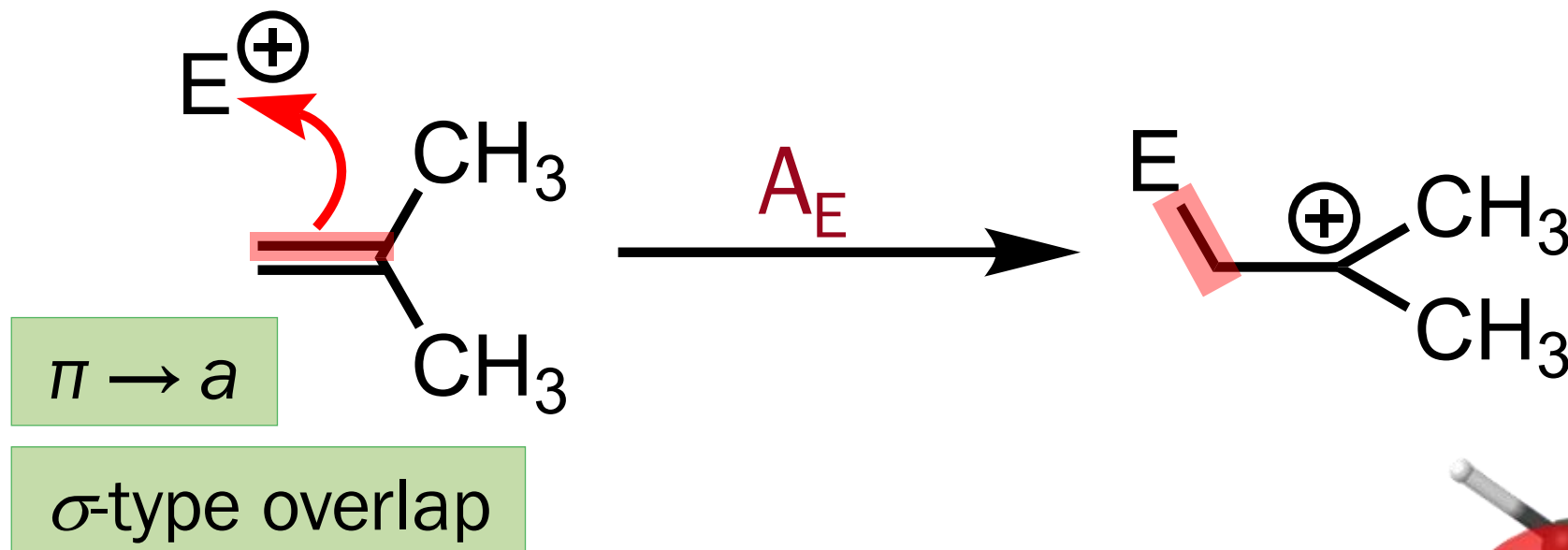


Curved arrows imply the localized molecular orbitals involved in an elementary step. In D_N , a sigma bond breaks, with both electrons going to the leaving group (nucleofuge). The electrons in the σ bonding orbital are accepted by an empty atomic orbital (a) on the leaving group. The charge of the leaving group increases by 1 and the charge of the other atom in the bond decreases by 1.

Electrophile Association (A_E)



Addition Reactions 1

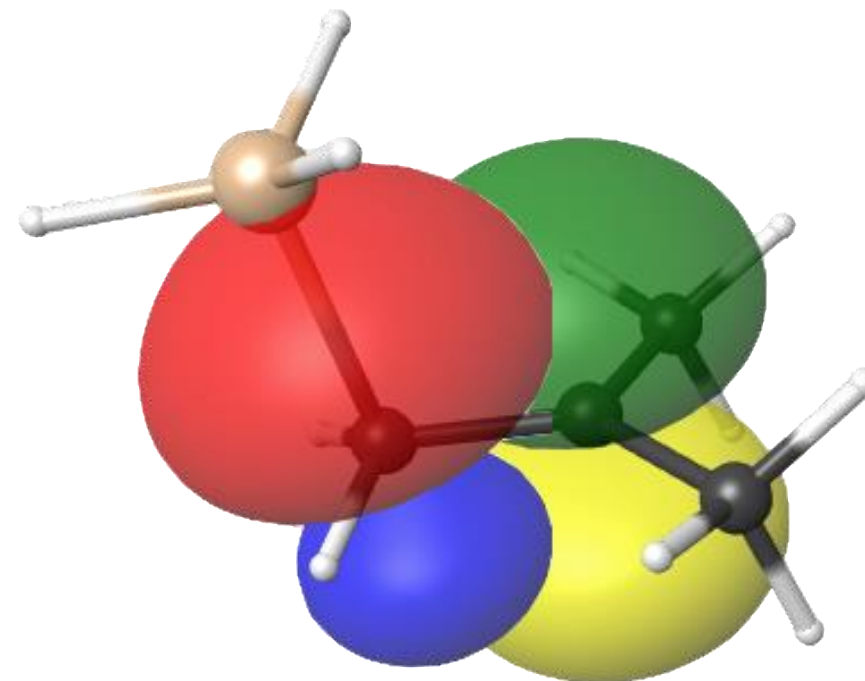
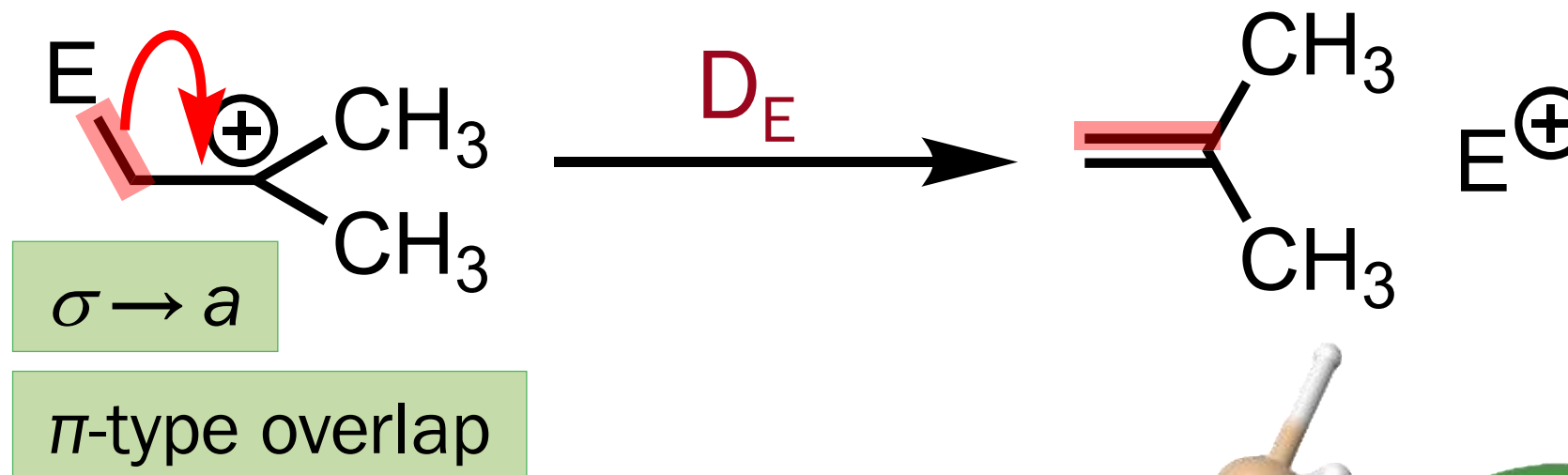


Curved arrows imply the localized molecular orbitals involved in an elementary step. In A_E , a pi bonding orbital (π) overlaps with an empty orbital (a) on an electrophile in a head-on, σ -type fashion. A new sigma bond is formed between the electrophile and one of the atoms of the pi bond. The charge of the other atom in the pi bond increases by 1 and the charge of E decreases by 1.

Electrofuge Dissociation (D_E)



Elimination Reactions 2

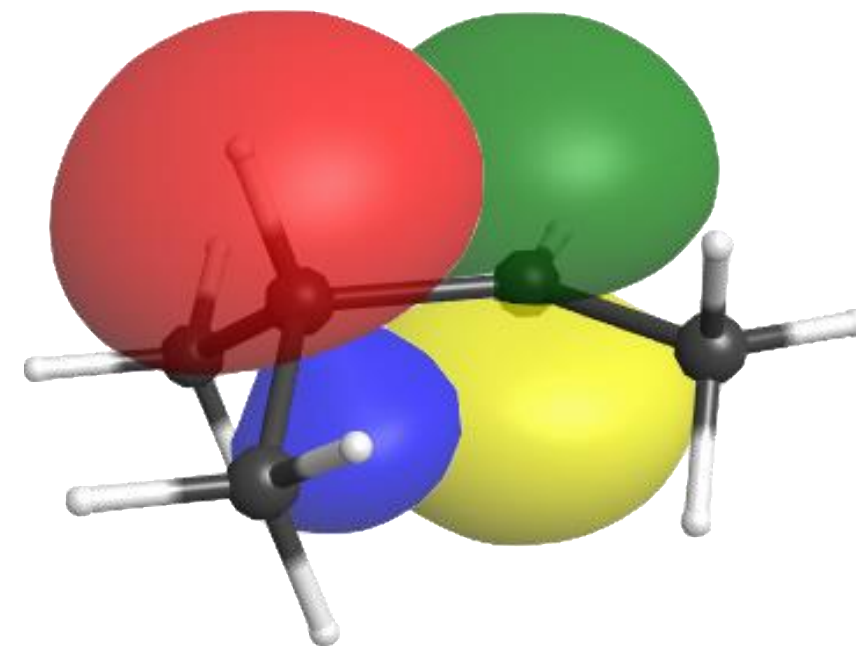
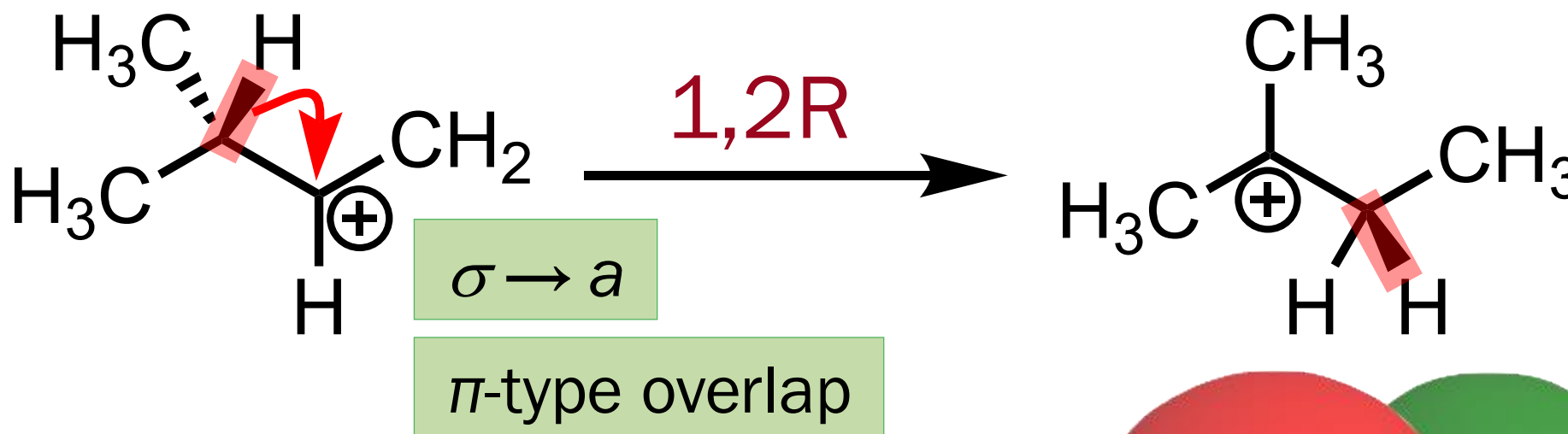


Curved arrows imply the localized molecular orbitals involved in an elementary step. In D_E , a sigma bonding orbital (σ) overlaps with an empty $2p$ orbital on an adjacent carbocation (a) in a side-on fashion. The $X-E$ electrons become part of a new pi bond and E departs as a positive leaving group (electrofuge). The atom between E and the carbocation need not be carbon.

1,2-Rearrangement (1,2R)



Substitution Reactions 17

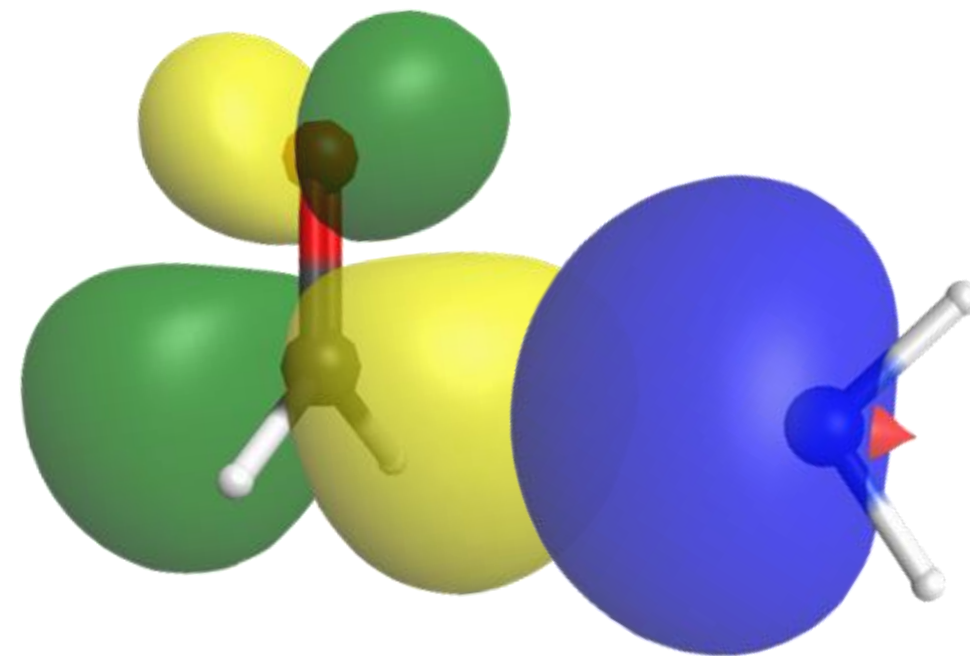
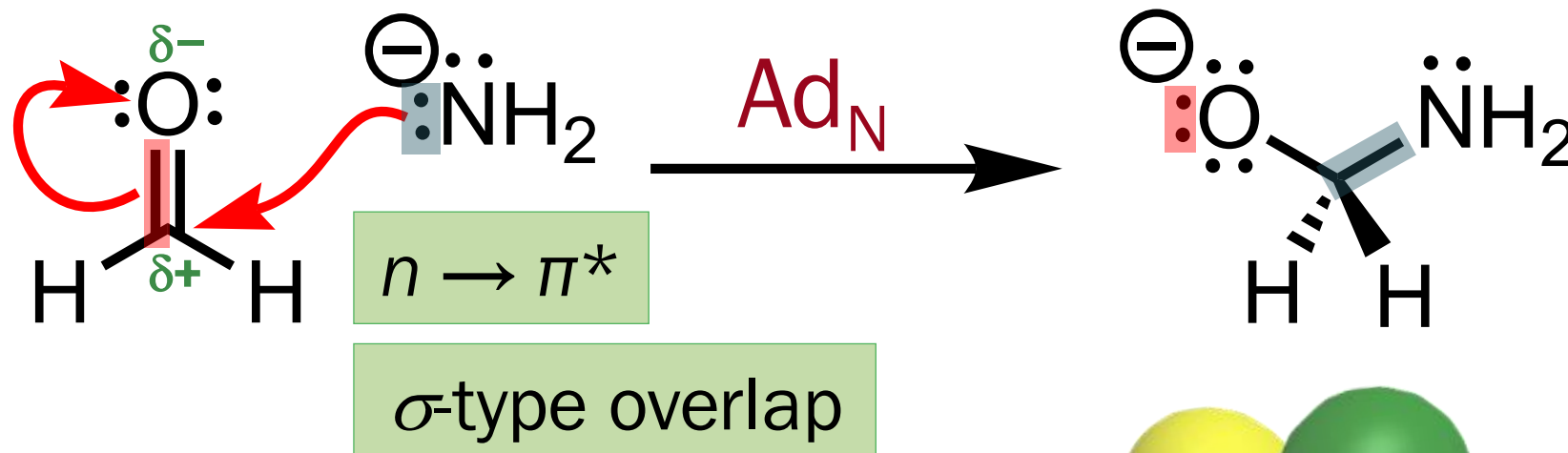


Curved arrows imply the localized molecular orbitals involved in an elementary step. In 1,2R, a sigma bonding orbital (σ) overlaps with an adjacent empty $2p$ orbital (a). The sigma bond migrates, carrying one of its atoms with it (usually an alkyl group or $-H$) and leaving the other atom with an increased charge. This is similar to D_E , except that the sigma bond migrates instead of breaking.

Nucleophilic Addition (Ad_N)



Carbonyl Reactions 2

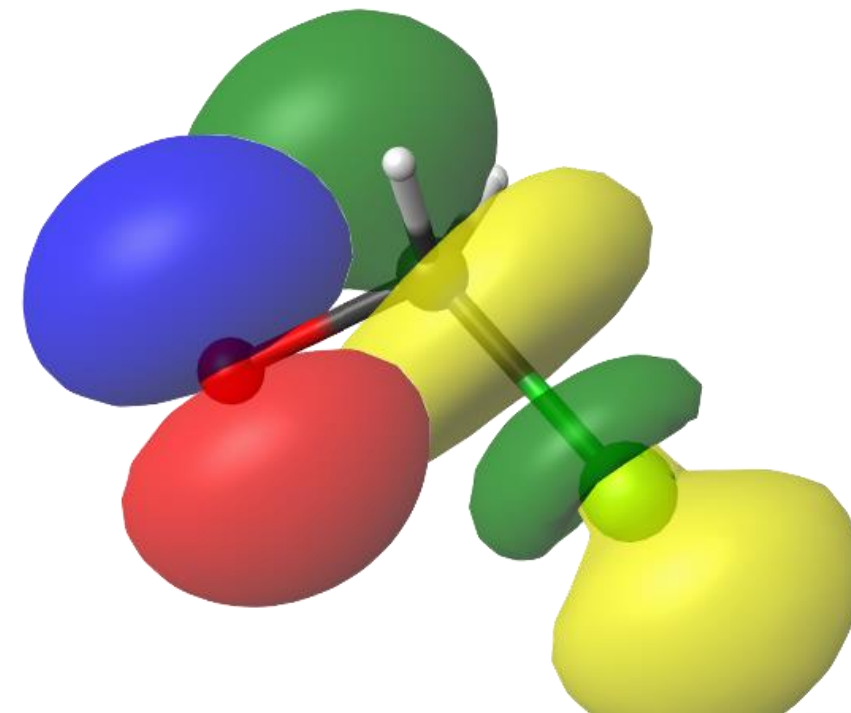
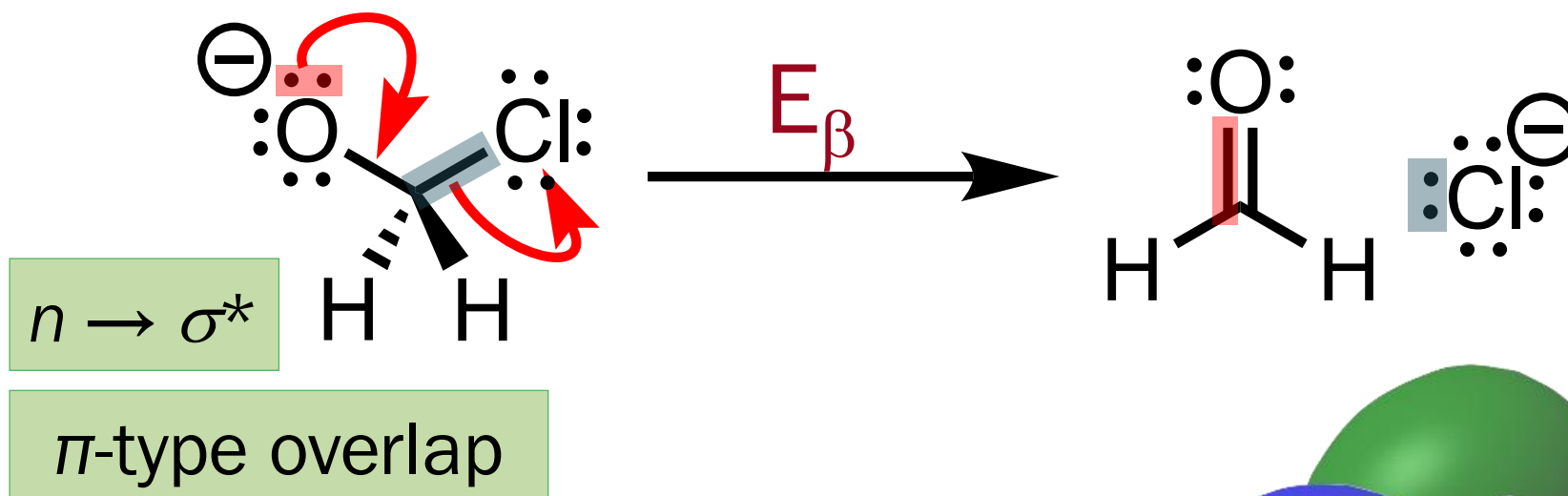


Curved arrows imply the localized molecular orbitals involved in an elementary step. In Ad_N , a nonbonding lone pair orbital (n) in the nucleophile overlaps in a head-on, σ -type fashion with the empty π^* orbital of a polarized pi bond in the electrophile, as suggested by cleavage of the pi bond. A new sigma bond forms and the electrons of the pi bond end up on the more electronegative atom.

Beta-elimination (E_β)



Carboxylic Acid Reactions 3

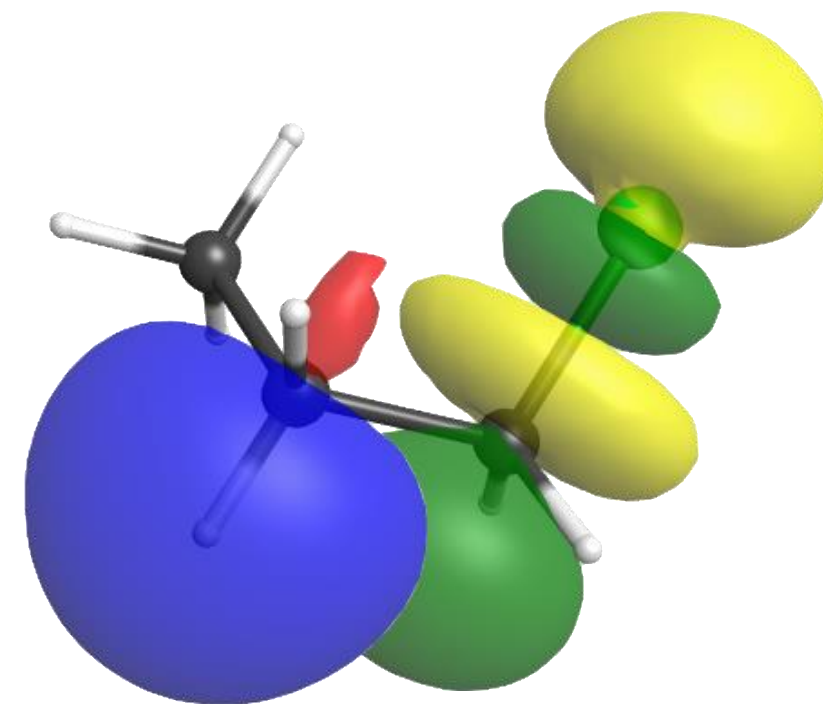
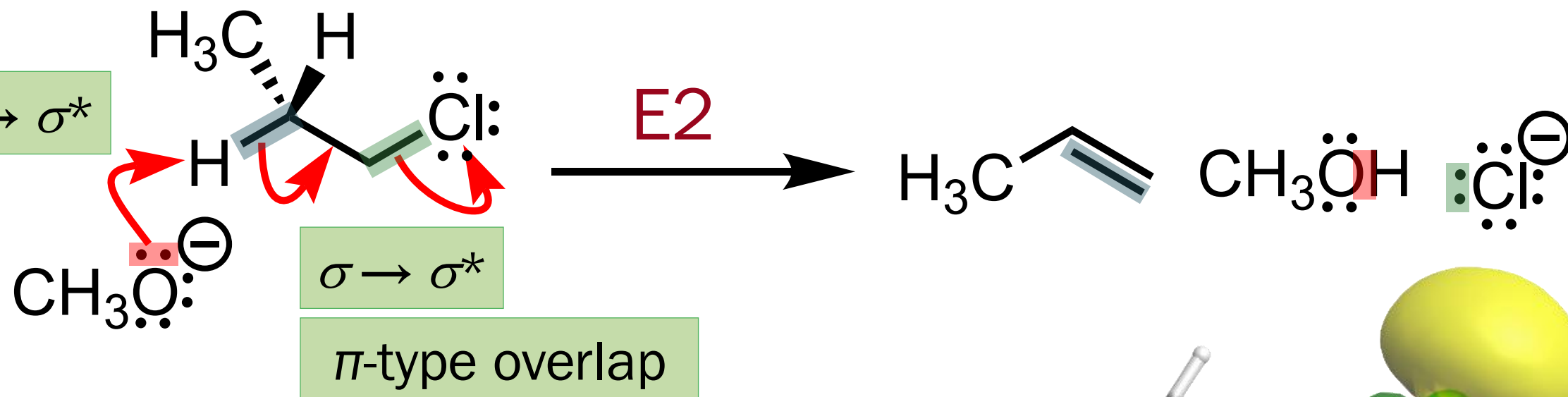


Curved arrows imply the localized molecular orbitals involved in an elementary step. In E_β , a nonbonding lone pair orbital (n) overlaps with the sigma antibonding orbital (σ^*) of an adjacent polarized bond in a side-on, π -type fashion. The polarized sigma bond breaks and the leaving group (typically an electronegative or positively charged atom) takes both electrons with it.

Bimolecular Elimination (E2)



Elimination Reactions 1

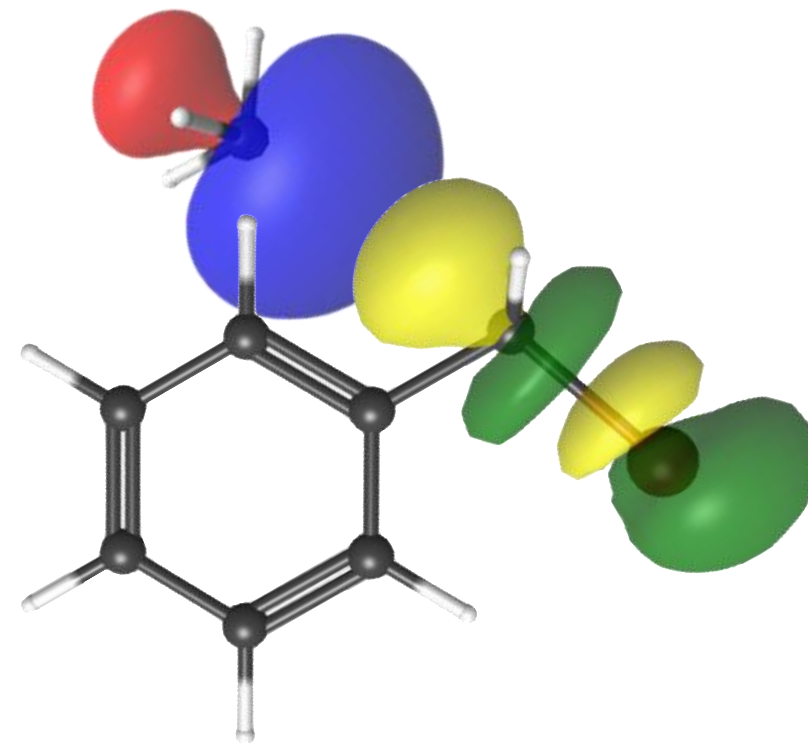
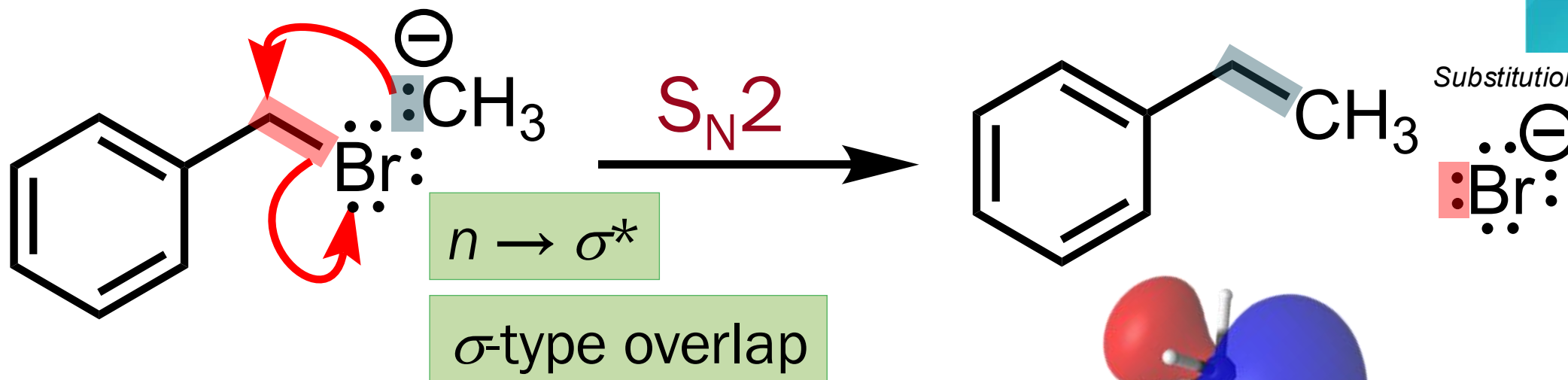


Curved arrows imply the localized molecular orbitals involved in an elementary step. In E2, a sigma bonding orbital (σ) overlaps with the sigma antibonding orbital (σ^*) of an adjacent polarized bond in a side-on, π -type fashion. The polarized sigma bond breaks and a new pi bond forms. The $\sigma \rightarrow \sigma^*$ interaction is promoted by a simultaneous $n \rightarrow \sigma^*$ interaction (proton transfer!).

Bimolecular Nucleophilic Substitution (S_N2)



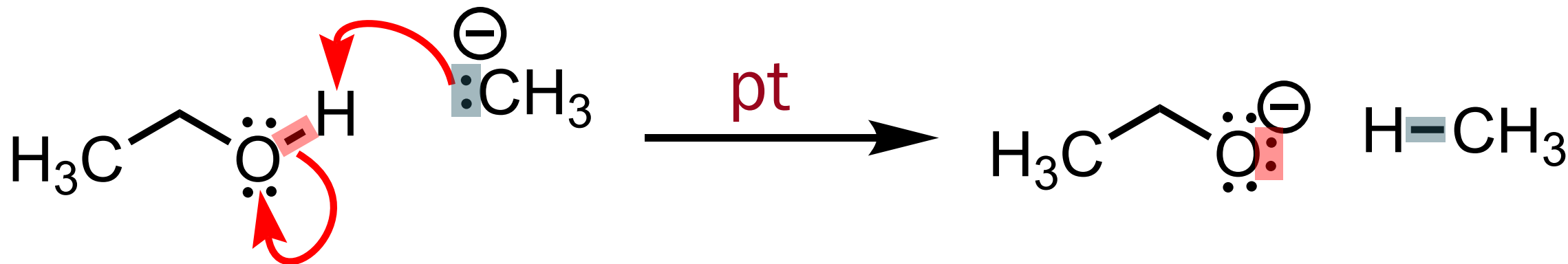
Substitution Reactions 1



Curved arrows imply the localized molecular orbitals involved in an elementary step. In S_N2 , a nonbonding lone pair on the nucleophile (Nu) overlaps with the σ^* orbital of the carbon-leaving group bond (C-X) in the electrophile. A new C-Nu bond forms and the C-X bond breaks, with the electrons in this bond going to X. Cleavage of the C-X bond implies the involvement of the σ^*_{C-X} orbital.

Acids and Bases in Organic Chemistry

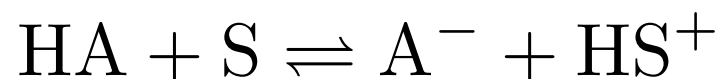
- Brønsted acid-base processes (pt) are extremely common in organic reaction mechanisms
 - Critical steps in acid- and base-catalyzed mechanisms
 - Can be an undesirable side reaction in some circumstances
- Proton transfer is mechanistically simple and is always governed by thermodynamics (rather than kinetics)



The Acid Dissociation Constant

The acid dissociation constant (K_a) is the equilibrium constant for the reaction of an acid with a given solvent, typically either water or dimethyl sulfoxide (DMSO).

- K_a reflects the balance of conjugate acid and base at equilibrium
- All Brønsted acid-base reactions are at equilibrium in organic reactions. **ALWAYS.**



$$K_a = \frac{[\text{A}^-][\text{HS}^+]}{[\text{HA}]}$$

$\text{p}K_a$ is more “human friendly” than K_a .

$$\text{p}K_a = -\log_{10} K_a$$

Recall that the strength of an acid (HA) and its conjugate base (A^-) are inversely related.

Survey of pK_a Values (Approximate)

Compound	pK_a (in H_2O)
H_3O^+	0
CH_3CO_2H	5
C_6H_5OH , HCN , HNR_3^+	10
ROH , H_2O	15
$RC(O)CH_3$	20
$RCCH$	25
NH_3 , H_2	35
C_6H_6	40
CH_4 , $H_2C=CH_2$	50

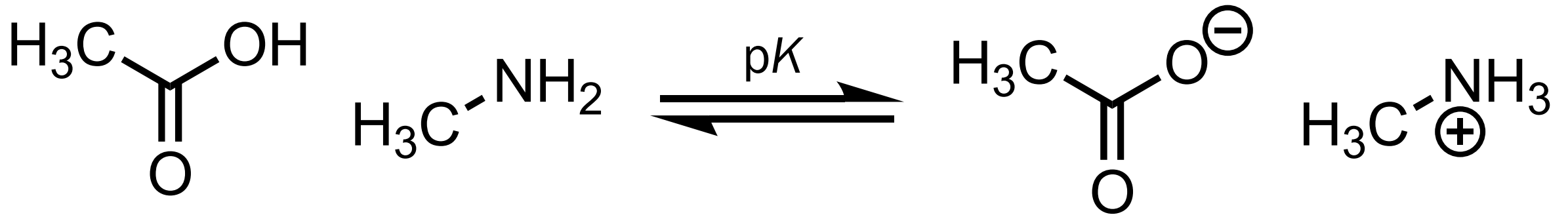
Use these as *benchmarks* and the structural factors affecting acidity (see below) to determine pK_a values relative to these.

Additional pK_a values are available [here](#).

Calculating pK | 1 of 2

Acetic acid undergoes deprotonation in this direction...

$pK_a(\text{CH}_3\text{CO}_2\text{H})$



Methylammonium undergoes deprotonation in this direction.

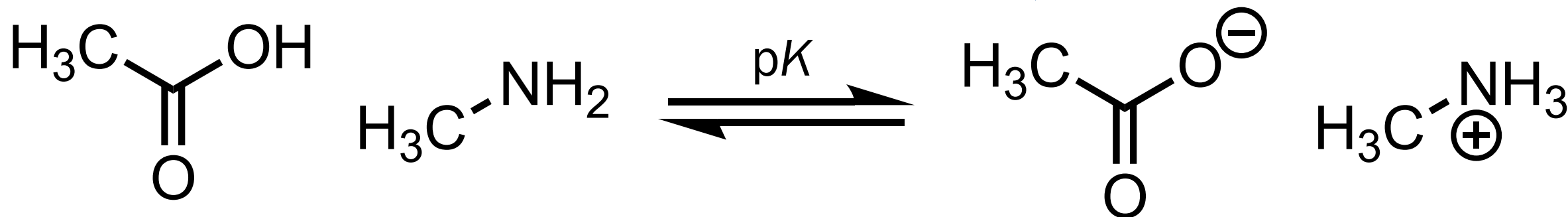
$-pK_a(\text{CH}_3\text{NH}_3^+)$

$$pK = pK_a(\text{CH}_3\text{CO}_2\text{H}) - pK_a(\text{CH}_3\text{NH}_3^+)$$

Calculating pK | 2 of 2

Acetic acid undergoes deprotonation in this direction...

$$pK_a(\text{CH}_3\text{CO}_2\text{H}) = 5$$



Methylammonium undergoes deprotonation in this direction.

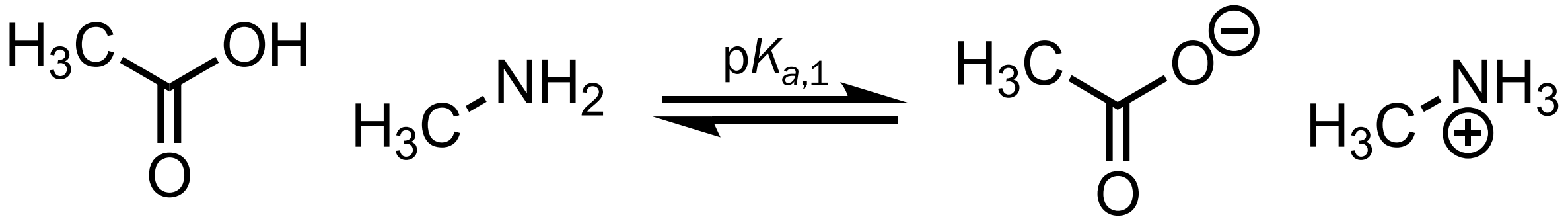
$$-pK_a(\text{CH}_3\text{NH}_3^+) = -10$$

$$pK = 5 - 10 = -5$$

The negative pK indicates that this reaction is heavily favored in the forward direction. The reaction will proceed forward as written.

Estimating pK for any Acid-base Reaction

The favored side of *any* acid-base reaction is the side with the weaker acid and base.



Acetic acid is a relatively strong acid. Its conjugate base is resonance stabilized.

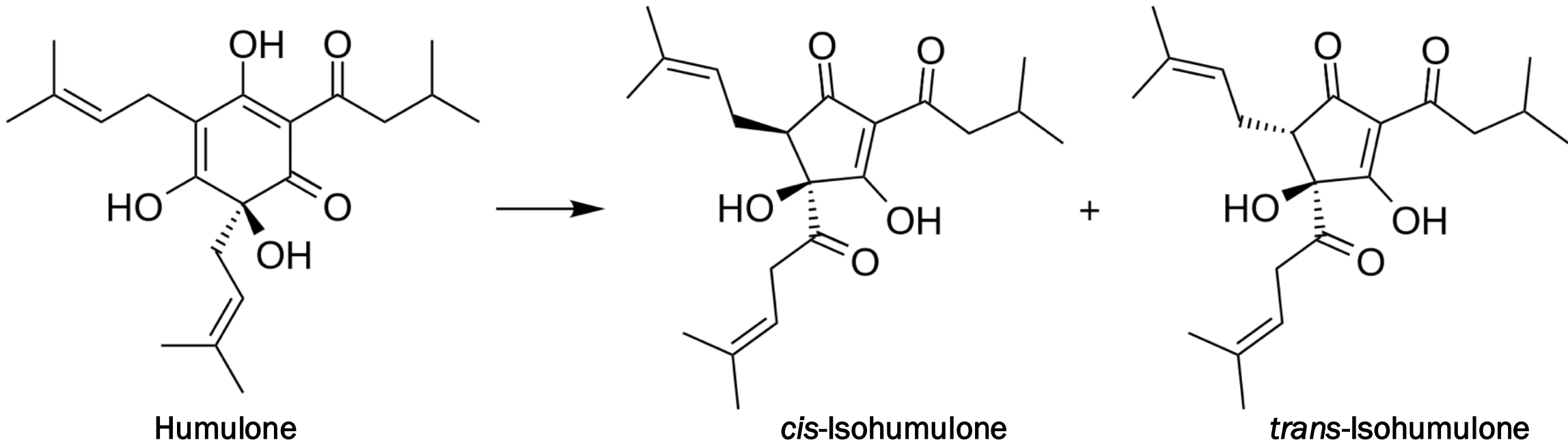
Methylammonium is a relatively weak acid. It bears positive charge on a less electronegative nitrogen atom.

Methylamine is a relatively strong base. Nitrogen's lone pair is readily donated to acetic acid.

Acetate is a relatively weak base. It is resonance stabilized and has negative charge on oxygen.

Stability Factors | 1 of 2

Stability factor: a structural *difference* between two molecules that causes a difference in stability (thermodynamic free energy) between the two.



Prolonged heating of humulone in hops causes isomerization to *cis*- and *trans*-isohumulone. Why are the products of this reaction more stable than the reactants? **Stability factors** guide our thinking.

Stability Factors | 2 of 2

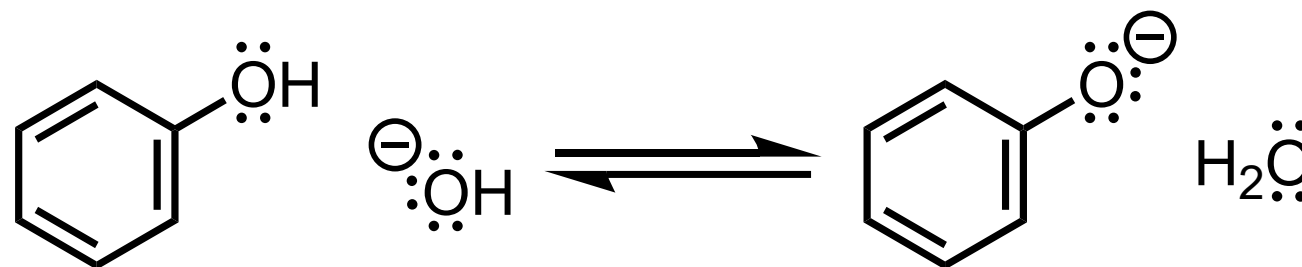
Stability factor: a structural *difference* between two molecules that causes a difference in stability (thermodynamic free energy) between the two.

- Five stability factors in order of importance
 - Electronegativity or size of charge-bearing atoms
 - Resonance delocalization
 - Hybridization
 - Inductive effects
 - Steric effects

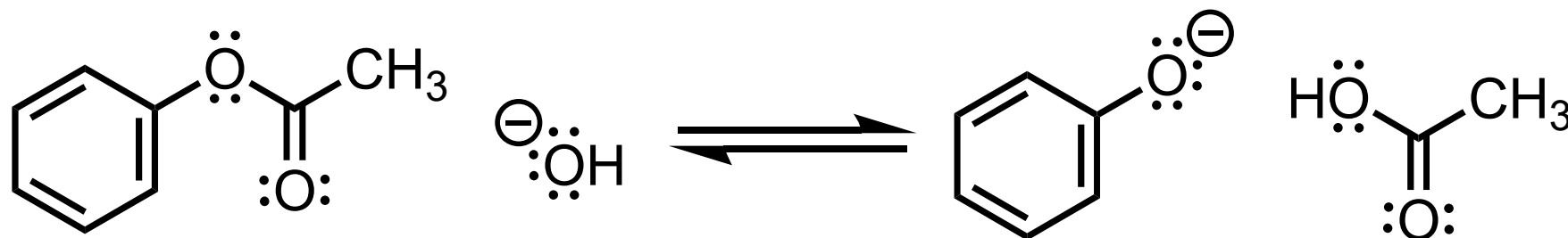
Focus on the Charged Species!

Focus on the charged species! Ions are often the least stable species in reactions and are more strongly affected by structural differences than neutral molecules.

Which side of this reaction is favored? Where do we look?

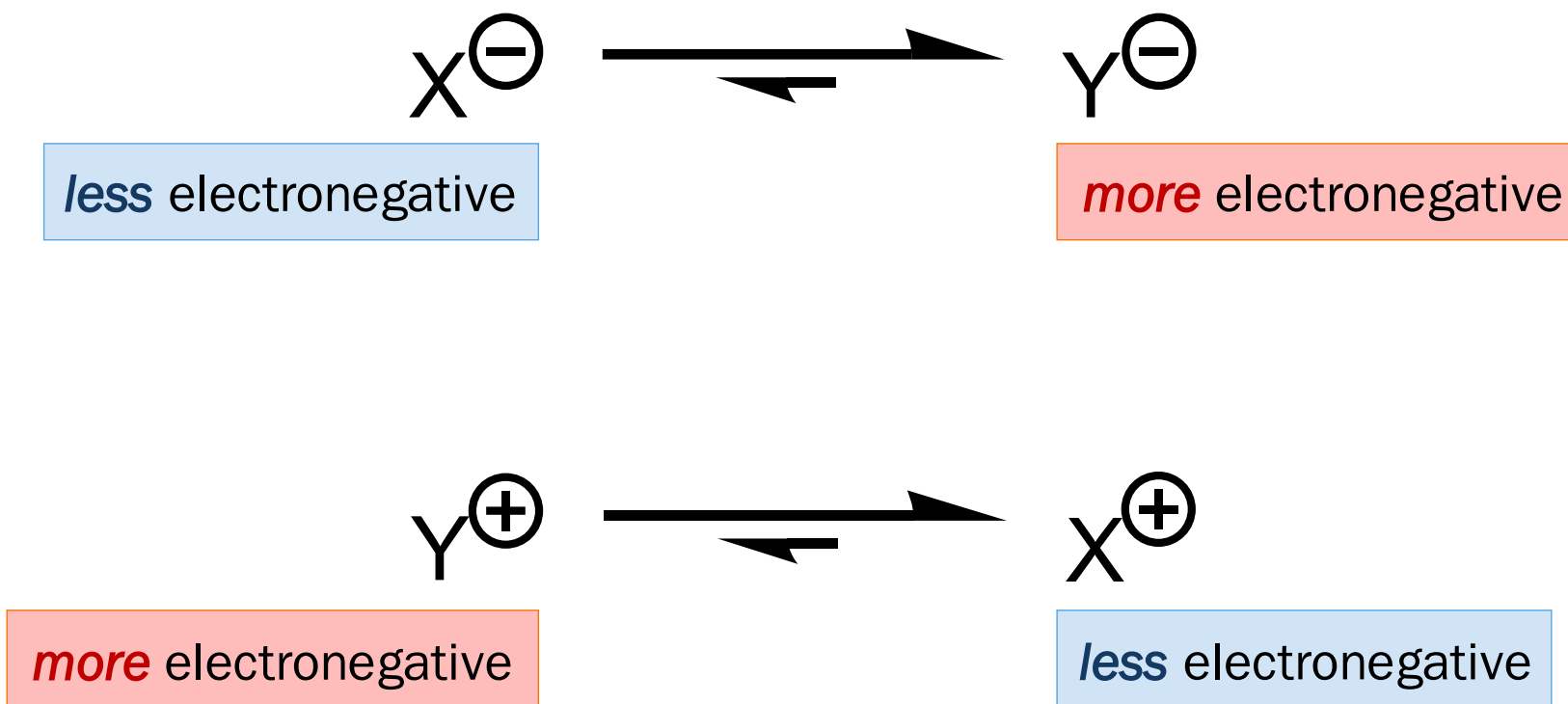


This type of thinking is generalizable...



Structural Determinants of Stability | 1 of 7

Electronegativity of charge-bearing atoms. Negative charge prefers to reside on more electronegative atoms. Positive charge prefers to reside on less electronegative atoms.

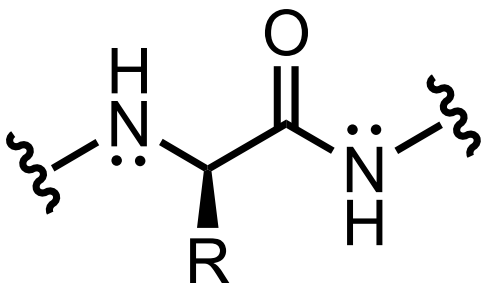


Structural Determinants of Stability | 2 of 7

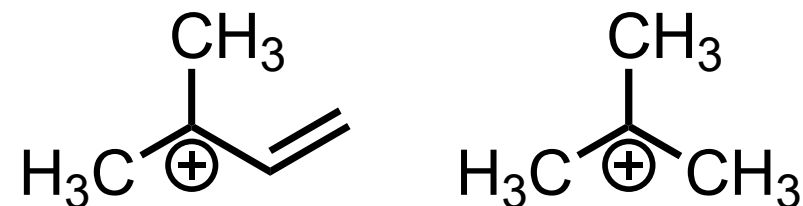
Resonance delocalization. A charged atom or lone pair engaged in resonance is more stable (less reactive) than one not engaged in resonance.

Consider the competition between internal and external electron donation/acceptance.

Which nitrogen is a stronger base?



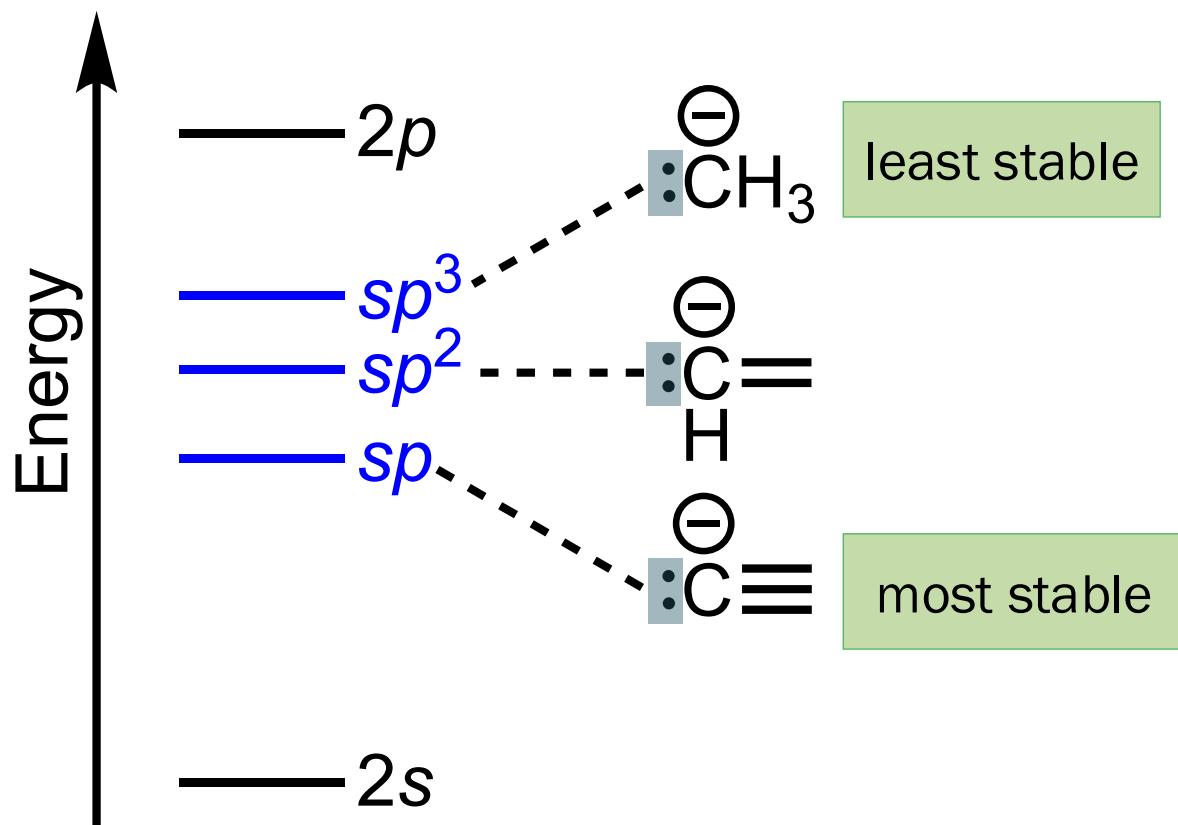
Which carbocation is more reactive as a Lewis acid?



Structural Determinants of Stability | 3 of 7

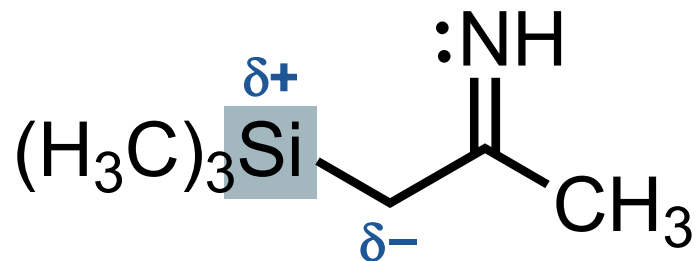
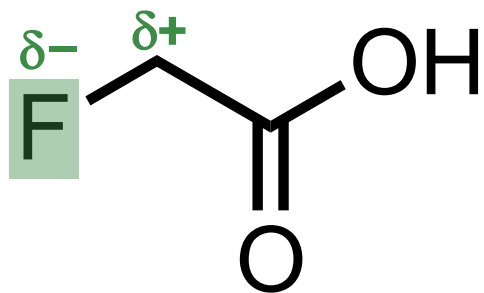
Hybridization. A lone pair occupying a hybrid orbital (i.e., not engaged in resonance) increases in stability as the ***s character*** of the hybrid increases.

Which anion is the strongest base?



Structural Determinants of Stability | 4 of 7

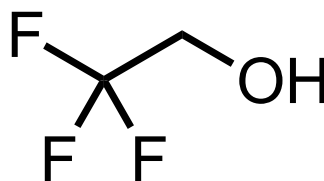
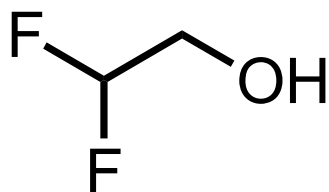
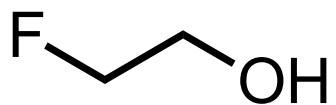
Highly electronegative or electropositive atoms withdraw or donate electrons through sigma bonds. The resulting effect on distant partial charges is called an inductive effect.



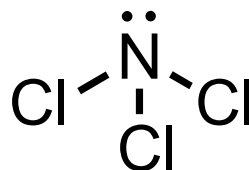
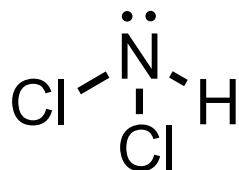
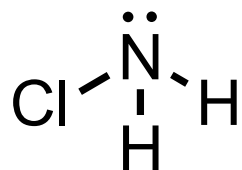
Structural Determinants of Stability | 5 of 7

Inductive effects. Inductively withdrawing atoms stabilize negative charges and destabilize positive charges. Inductively donating atoms stabilize positive charges and destabilize negative charges.

Which alcohol is the strongest acid?



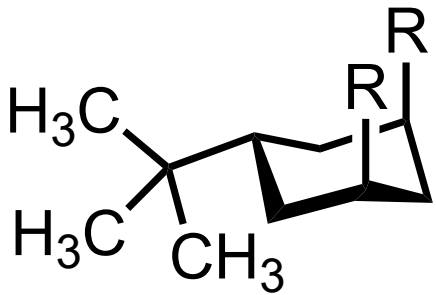
Which amine is the strongest base?



Structural Determinants of Stability | 6 of 7

The term **steric** refers to the spatial environment around a group, especially emphasizing the *accessibility* of other molecules to the group as a result of nearby electron clouds.

Groups “bumping” into one another as a result of the forced shape of a molecule engage in **steric repulsion**.



An atom/group surrounded by many other atoms is said to be **sterically crowded** or **hindered**.

Structural Determinants of Stability | 7 of 7

Steric effects. Forced steric repulsion is always destabilizing. However, steric hindrance can have a stabilizing or destabilizing effect depending on the situation.

Conversion of less hindered to more hindered structures is disfavored.

Hindered atoms may be shielded or protected from reaction.

Solvation of sterically hindered charged atoms is often difficult.